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In [1]:
```
import pandas as pd
import numpy as np
data = {
    'Rank': [1, 3, 2, 4, 6, 5, 7, 8, 9, 10, 12, 11, 13, 15, 14, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28,
    'City': ['Tampa', 'Singapore', 'Breda', 'Oslo', 'Vancouver', 'Sydney', 'Montreal', 'Cambridge', 'Xalapa', 'Durb
             'Sacramento', 'Johannesburg', 'Frankfurt', 'Veracruz', 'Geneva', 'Amsterdam', 'Seattle', 'Toronto', 'M
             'Boston', 'Tel Aviv', 'Guadalajara', 'León', 'Turin', 'Los Angeles', 'Buenos Aires', 'Franca', 'New Yo
             'Cape Town', 'London', 'São Paulo', 'Quito', 'Kobe', 'Paris'],
    'Country_Code': ['USA', 'SGP', 'NLD', 'NOR', 'CAN', 'AUS', 'CAN', 'USA', 'MEX', 'ZAF',
                     'USA', 'ZAF', 'DEU', 'MEX', 'CHE', 'NLD', 'USA', 'CAN', 'USA',
                     'USA', 'ISR', 'MEX', 'MEX', 'ITA', 'USA', 'ARG', 'BRA', 'USA',
                     'ZAF', 'GBR', 'BRA', 'ECU', 'JPN', 'FRA'],
    'GVI': [36.1, 29.3, 29.3, 28.8, 25.9, 25.9, 25.5, 25.3, 24.0, 23.7,
            23.6, 23.6, 21.5, 21.4, 21.4, 20.6, 20.0, 19.5, 19.4,
            18.2, 17.5, 17.3, 16.5, 16.2, 15.2, 14.5, 13.7, 13.5,
            13.4, 12.7, 11.7, 10.8, 9.4, 8.8]
}

df = pd.DataFrame(data)
```

1. Average of GVI of "USA"

In [11]:
```
avg_usa = df[df['Country_Code'] == 'USA']['GVI'].mean()
print(f" Average GVI of USA: ",avg_usa)
```

```
Average GVI of USA:  21.4125
```

2. Minimum of GVI of "Netherlands"

In [12]:
```
min_nld = df[df['Country_Code'] == 'NLD']['GVI'].min()
print(f"Minimum GVI of Netherlands: ", min_nld)
```

```
Minimum GVI of Netherlands:  20.6
```

3. Sum of GVI ("Cities of Italy")

In [13]:
```
sum_ita = df[df['Country_Code'] == 'ITA']['GVI'].sum()
print(f" Sum of GVI for Italy: ",sum_ita)
```

```
Sum of GVI for Italy:  16.2
```

4. Find the city and country with least GVI

In [44]:
```
res = df.groupby(['Country_Code', 'City'])['GVI'].agg('min')
print(res.idxmin()[0], res.idxmin()[1], res.min())
```

```
FRA Paris 8.8
```

5. Find Countries which has maximum number of cities in top 30

In [45]:
```
counts = df[df['Rank'] <= 30].groupby('Country_Code')['City'].agg('count')
print(f"Top Country: {counts.idxmax()} ({counts.max()} cities)")
```

```
Top Country: USA (8 cities)
```

6. Summary

a) df.describe()

In [49]:
```
df.describe()  #statistical summary
```

Out[49]:

	Rank	GVI
count	34.000000	34.000000
mean	17.500000	19.829412
std	9.958246	6.434355
min	1.000000	8.800000
25%	9.250000	14.675000
50%	17.500000	19.750000
75%	25.750000	23.925000
max	34.000000	36.100000

b) pivot table

In [50]:
```
pivot_df = pd.pivot_table(
    df,
    values='GVI',
    index='Country_Code',
    aggfunc=['mean', 'min', 'max']
)
print(pivot_df)
```

Country_Code	mean GVI	min GVI	max GVI
ARG	14.500000	14.5	14.5
AUS	25.900000	25.9	25.9
BRA	12.700000	11.7	13.7
CAN	23.633333	19.5	25.9
CHE	21.400000	21.4	21.4
DEU	21.500000	21.5	21.5
ECU	10.800000	10.8	10.8
FRA	8.800000	8.8	8.8
GBR	12.700000	12.7	12.7
ISR	17.500000	17.5	17.5
ITA	16.200000	16.2	16.2
JPN	9.400000	9.4	9.4
MEX	19.800000	16.5	24.0
NLD	24.950000	20.6	29.3
NOR	28.800000	28.8	28.8
SGP	29.300000	29.3	29.3
USA	21.412500	13.5	36.1
ZAF	20.233333	13.4	23.7

c) Using groupby and aggregation function

In [51]:
```
summary_df = df.groupby('Country_Code')['GVI'].agg(['mean', 'min', 'max'])
print(summary_df)
```

Country_Code	mean	min	max
ARG	14.500000	14.5	14.5
AUS	25.900000	25.9	25.9
BRA	12.700000	11.7	13.7
CAN	23.633333	19.5	25.9
CHE	21.400000	21.4	21.4
DEU	21.500000	21.5	21.5
ECU	10.800000	10.8	10.8
FRA	8.800000	8.8	8.8
GBR	12.700000	12.7	12.7
ISR	17.500000	17.5	17.5
ITA	16.200000	16.2	16.2
JPN	9.400000	9.4	9.4
MEX	19.800000	16.5	24.0
NLD	24.950000	20.6	29.3
NOR	28.800000	28.8	28.8
SGP	29.300000	29.3	29.3
USA	21.412500	13.5	36.1
ZAF	20.233333	13.4	23.7

In [ ]: